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CURRENT OPINION

Is It All in Your Head? Placebo Effects in Concussion Prevention

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Abstract
Sports-related concussions (SRCs) are a major health issue in athletes of varying ages, ability levels, and sports. Concerns over the short- and long-term consequences of SRCs have incentivized a wealth of products and policies aimed at reducing SRC risk. Research suggesting the effectiveness of such interventions at reducing SRCs has facilitated their adoption by sports organizations and, in some cases, product commercialization. However, the body of SRC mitigation research is almost entirely devoid of placebo or sham groups, which raises important questions about the true clinical effectiveness of these interventions. This Current Opinion explores the plausibility of placebo effects within the scope of SRC prevention, describes why the lack of placebo/sham groups in the current body of literature is problematic, and provides recommendations for including placebo/sham groups in future SRC research.

Key Points

Research examining the effectiveness of interventions designed to reduce the risk of sports-related concussions is universally conducted without blinding participants to the assigned intervention, or the inclusion of a placebo/sham group.

Sports-related concussion diagnosis is heavily dependent on honest self-reporting of symptoms, and is likely to be sensitive to placebo effects.

Future research examining sports-related concussion mitigation should include placebo/sham groups to better understand the true effectiveness of the interventions of interest.

1 Introduction

Sport-related concussions (SRCs) are a persistent issue across all contact sports, at all levels of play. This has created a need for interventions to reduce SRC risk. In a recent systematic review, strategies such as protective devices (e.g. helmets, mouthguards), changes to sports policy/laws, or changes to practice conditions (e.g., contact levels, tacking techniques) were each considered effective for mitigating SRC risk [1]. These interventions were also linked with athletes reporting a decrease in both the severity and duration of SRC symptoms [1]. Likewise, users of SRC prevention products also report a strong belief in their effectiveness. However, there is a glaring oversight in the observational and experimental research in this field—a possible placebo effect. In this Current Opinion, we discuss why the placebo effect must be stringently examined in SRC research and provide recommendations for characterizing and quantifying its response in future empirical studies.

2 Overview of Placebo and Nocebo Effects

An observed clinical effect represents the aggregate of specific treatment effects and contextual effects. The latter encompasses both natural recovery and placebo effects, which are changes specifically attributable to placebo mechanisms (e.g., the neurobiological and psychological mechanisms of expectations) [2].

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The placebo effect refers to enhanced clinical outcomes (e.g., reduced symptom severity, improved function, faster recovery) due to positive expectations, whereas the nocebo effect is impaired clinical outcomes (e.g., increased symptom severity, reduced function, prolonged recovery) due to negative expectations. A 2018 consensus statement for sports researchers defines placebo and nocebo effects as desirable or undesirable outcomes, resulting from a person's expected and/or learned response to a treatment or situation [3].

Contextual effects can have a significant impact on patient care. In a meta-review of 186 clinical trials using a range of pharmacological, physical, or psychological interventions, at least half of the overall treatment effect observed was attributable to contextual effects, rather than the specific experimental intervention on trial [4]. Similar effects are evident in sports research. For instance, a meta-analysis involving 32 controlled trials found that nutritional or mechanical ergogenic aids have small-to-moderate placebo effects on sporting performance [5].

2.1 Mechanisms of Placebo and Nocebo Effects

The physiological mechanisms of placebo and nocebo effects have been described extensively elsewhere [6–9], and details are beyond the scope of this paper. In brief, the placebo effect is primarily mediated by the release of endogenous opioids, dopamine, and other neurotransmitters that modulate pain perception, mood, and immune function. In contrast, the nocebo effect involves increased activation of the hypothalamic–pituitary–adrenal axis and elevated stress-related hormones (e.g., cortisol). Imaging studies suggest placebo responses are associated with activation of reward pathways and inhibition of the pain-, anxiety-, and fear-related pathways, whereas nocebo effects produce the opposite effect, although the mechanisms of each effect are far more nuanced.

The placebo response is a complex phenotype that is often heterogeneous within a population [10]. Some of the variation may relate to genetics [11], and a network-based genetic analysis concluded that at least 28 genes are involved in placebo–drug interactions [10]. However, few clinical studies have specifically aimed to disentangle genetic effects from behavioral and environmental mechanisms. In a seminal twin study examining placebo analgesia and its mechanisms, Weimer et al. [12] concluded that heritability was actually a weak predictor of placebo responsiveness, with most of the variation in response explained by the nature of the therapeutic intervention.

2.2 Interaction Effects

A further consideration is that active and contextual effects may not just be additive, they may also interact to either enhance (synergistic) or reduce (antagonistic) the effects of a given treatment [13]. Certain experimental conditions or environments may be more sensitive to large interactions, particularly when symptoms are perceived/self-reported. A study by Bingel et al. [14] exemplifies how individual expectations can alter the analgesic effects associated with a potent opioid (remifentanil); their reported interaction effects were large, with positive expectations doubling the drug's analgesic benefit (placebo), whereas negative expectations completely abolished its therapeutic effect (nocebo). Similar findings have been reported from randomized controlled trials (RCTs) in sports performance, whereby the nature of the placebo administration (giving athletes positive vs negative information) dictated the improvement (or not) of cycling performance [15].

2.3 Placebo Effects in Disease and Injury Prevention

Medical research regarding placebo effects is often focused on clinical management or treatment of an existing medical condition, rather than prevention of a new condition. There are rare exceptions to this in clinical settings (e.g., placebo vaccines to prevent infectious disease) [16]. Trials to examine whether a placebo effect can influence the pathogenesis of new disease would require a suitably powered sample size (dependent on the prior probability of participants developing the condition) and would need to consider relevant confounding factors. In the sports context, the placebo effect has been studied from a performance enhancement perspective, but its influence on injury prevention has been largely overlooked.

3 Placebo Versus Nocebo Effects in SRC

It is possible that athletes may experience SRC-related placebo effects or nocebo effects, depending on expectations.

3.1 The Role of Nocebo Effects in SRC

There is increasing evidence that concussion outcomes are affected by many psychosocial constructs, with depression, anxiety, and neuroticism associated with prolonged recovery. Polich et al. [17] and Madsen et al. [18] suggest that media coverage can foster nocebo effects in concussion outcomes and prognosis. They theorize that concussion awareness campaigns, which were originally

developed to address under-reporting, inadvertently lower recovery expectations, leading to worse clinical outcomes. Athletes who voluntarily adopt SRC prevention measures may be more concerned about SRC— this might promote negative expectations about SRC risk and recovery, and facilitate nocebo responses.

In a research setting, those assigned to the control group may be more pessimistic about their SRC risk if they believe the intervention group is more protected, and this may be amplified if they have heightened anxiety about SRC risk. These attitudes may influence SRC reporting intentions [19, 20], and possibly have physiological consequences (e.g., increased pain perception, increased cortisol production). Likewise, those assigned to SRC interventions in research settings may become more aware of their risks of SRC, or perhaps the dangers associated with brain injuries. All of these factors could theoretically increase sensitivity to SRC symptoms and their severity. Thus, a nocebo effect could distort SRC clinical trial outcomes. We speculate that this nocebo effect would be greatest in the control group, which would then exaggerate the between-group differences in an SRC intervention study (i.e., to favor the treatment condition).

3.2 SRC Interventions Are Likely Dominated by Placebo Effects

While nocebo effects do occur in SRC, we posit that placebo effects are likely to dominate responses to SRC interventions (Fig. 1). Patterns of nocebo effects are not currently evident in the SRC prevention literature (i.e., SRC interventions are rarely associated with increased SRC prevalence) [1]. The competitive and pressurized culture prevalent in athletic environments is likely more conducive to a placebo response than a nocebo response in the context of SRC. Athletes are frequently conditioned to prioritize participation, performance, and resilience, often at the expense of their well-being. Sport-related concussion diagnosis can result in lost playing time and increased risk of losing a starting position on a team, and have negative long-term career implications [21–23] (but there is conflicting research on these effects [24–26]). In professional sports, there is a "play through pain" culture,

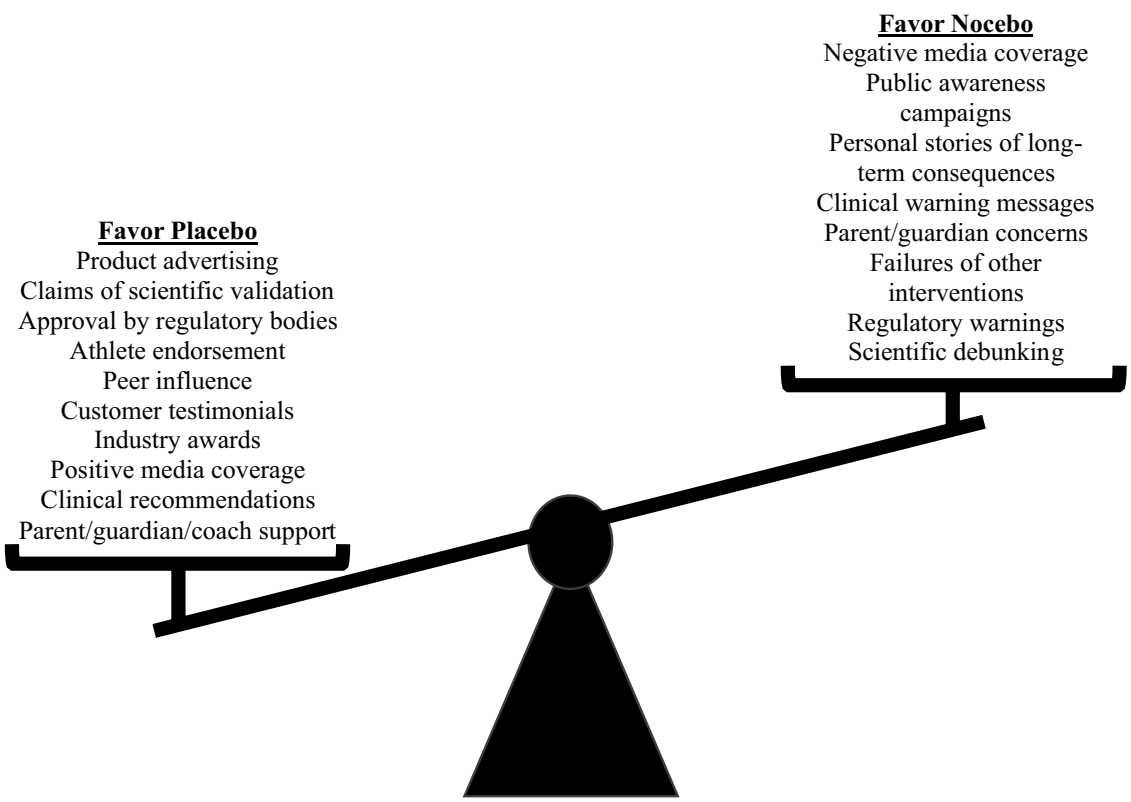

Fig. 1 Key factors influencing placebo and nocebo effects in sports-related concussion prevention. While the number of key factors is similar on both sides, individuals are likely more frequently exposed to placebo-supporting influences, such as daily interactions with advertisements and peers. Additionally, athletes who choose to participate in contact sports (or continue to participate following sports-related concussion) may exhibit selection bias having made personal risk decisions based on prior exposure to these factors. Thus, we suggest that placebo effects exert a far greater influence on responses to sports-related concussion prevention interventions than nocebo effects

where athletes will often intentionally mask injury symptoms (including SRC) and avoid medical attention [27]. Such risk-taking behavior also pervades adolescent sports [28]. It also seems plausible that athletes most influenced by nocebo effects withdraw from contact sports earlier in their career, thus leaving higher level athletes (e.g., college and professional) less susceptible to nocebo effects (survivorship bias).

There is a culture for stakeholders who acknowledge the risks of SRC to also promote interventions that claim to alleviate such risks. There are countless anecdotal examples of this. For instance, the National Football League (NFL) has repeatedly acknowledged that SRCs are a concern, and then regularly touted the effectiveness of its interventions to address this issue [29]. Prominent healthcare providers who have created awareness over chronic traumatic encephalopathy (CTE) risks have also endorsed products claiming to improve brain health and mitigate risk [30]. Likewise, athletes who speak about the potential dangers of concussions also evangelize products touting "brain protection" [31–33]. Thus, stakeholders at the individual and organizational levels, which raise awareness of negative expectations (and thus promote a nocebo effect), sometimes attenuate these fears by "proposing" solutions that create positive expectations (which promote a placebo effect). In Sects. 4 and 5, we describe the constellation of factors that athletes are regularly exposed to, which raise expectations that SRC preventions will be effective, and thus favor a placebo effect (over nocebo effect) for SRC interventions, for both individual-level perceptions and in research outcomes.

4 SRC-Specific Factors That May Promote a Placebo Effect

It has been proposed that placebo effects may have a strong influence in post-concussion recovery [34], and evidence of this is just beginning to emerge [35]. Likewise, it has been suggested that nocebo effects may be common in exacerbating symptoms of pediatric concussions [17]. However, no research has examined placebo effects in SRC prevention.

There are many reasons why experimental research into SRC mitigation will be placebo or nocebo sensitive. The primary reasons are: (1) issues with accurately diagnosing and reporting concussion, and (2) athlete and stakeholder (e.g., coaches, parents, medical staff) expectations for SRC-related interventions.

4.1 Concussion Diagnosis is Rooted in Clinical Symptoms

There is no single gold-standard test for concussion diagnosis and recovery, meaning that clinicians depend on clinical signs and patient-reported symptoms, rather than objective biomarkers [36–39]. Without an objective diagnostic test (e.g., blood or imaging biomarker with high sensitivity and specificity for mild traumatic brain injury), there is no foolproof method to determine if an athlete who is not displaying visible signs of SRC (e.g., disorientation, slow to get up, ataxia, grabbing head) does not actually have symptoms (true negative) or is simply not reporting them (false negative). Research has demonstrated changes in numerous physiological outcomes following a concussion (e.g., pupillary light response, evoked potentials); however, these data typically show a substantial inter-individual variation, and they have not yet been developed into clinically validated diagnostic measures. Similarly, while cognitive tests are promoted for use, in conjunction with clinical screening [40], there are related concerns around inter-individual variations and learning effects, which affect their test–retest reliability and diagnostic accuracy [37, 41].

Consequently, current guidelines are that concussion is diagnosed primarily on physical signs and clinical symptoms associated with an observable impact [37–39]. However, the "constellation" of signs and symptoms, and the disparities in how athletes present, if at all, makes concussion diagnosis challenging. The heavy reliance on symptom reporting when making a concussion diagnosis creates a conflict of interest for the athlete. The desire to continue playing, or the competitive pressure from peers or coaches can incentivize an athlete to under-report or mask their SRC symptoms [42]. Others may have received less education on concussion, leading to a lack of awareness of its signs, symptoms, and potential dangers.

4.2 Concussion Symptoms Often Go Unreported

When a SRC diagnosis is heavily or entirely dependent on honest symptom reporting from the athlete, diagnosis does not always occur—a well-described phenomenon [43–46]. A systematic review of 12 studies found 40–63.6% of athletes continued to participate in sports activities with concussion symptoms [47]. Such under-reporting means concussion rates are likely to be much higher [48]. These issues are not indicators of placebo effects themselves, but rather demonstrates some of the shortcomings when diagnosis is underpinned by symptom reporting, and how external factors can affect diagnostic validity/accuracy in athletic populations.

While exact data on the proportion of SRCs that are missed by "concussion spotters" or team doctors are unavailable, retrospective athlete surveys suggest that 33–50% sustained an impact to the head causing signs and/or symptoms of concussion, but were not clinically diagnosed [49, 50]. These surveys are limited by their retrospective design and recall of past concussion events, but a heavy reliance on patient symptoms in SRC diagnosis makes a

placebo effect very likely. The role of a symptom-based diagnosis in placebo effects is further explored in Sect. 6, and the limitations of current SRC mitigation research is discussed in Sect. 8.

4.3 Delayed Symptoms May Introduce Reporting Bias

A further caveat is that concussion symptoms can be delayed, with an onset that ranged from hours to days after the causative event. When athletes do not immediately exhibit symptoms following a head impact, there may be a window for false reassurance (optimism bias), potentially leading them to dismiss the possibility of a concussion, especially if they were employing a SRC mitigation strategy at the time of the impact. Indeed, athletes who are optimistic that they will not sustain an SRC have a lower intention to report symptoms compared with athletes who believe they are more likely to experience an SRC [19, 20]. Anchoring bias, whereby an individual relies too heavily on the first piece of information encountered (e.g.. head trauma with no immediate symptoms, passing an initial on-field concussion screen) or attribution bias (e.g., believing pre-existing conditions are the cause of SRC symptoms) can be problematic in healthcare decision making, and can affect diagnostic accuracy [51–53].

4.4 Performance and Detection Bias in SRC Research

Bias in experimental research can lead to systematic error, where an intervention's effect is either over- or underestimated. Masking (blinding) is essential to ensure that participants have a similar research experience pertaining to their clinical interactions, ancillary treatment, and diagnostic examination [54], particularly if subjective outcome reporting is used. Study design characteristics can have a significant impact on effect estimates; for example, a lack of blinding or unclear blinding in randomized trials exaggerates treatment effects (based on odds ratios) by an average of 13–36% [55, 56].

A recent systematic review shows that the SRC prevention literature usually lacks blinding and is not designed to examine the possibility of a placebo effect (details provided in Sects. 8 and 9). Thus, it is possible that athletes in SRC intervention studies are: (1) intentionally not reporting symptoms (acute or delayed) that they attribute to concussion; (2) experiencing a placebo effect that minimizes symptoms; (3) misattributing SRC symptoms to another causative factor (e.g., "unrelated" migraine, stress, dehydration, fatigue); or (4) actually protected against concussion (Fig. 2).

5 Social Factors That Influence Athlete Expectations in SRC Prevention

Sport is a unique environment owing to its ubiquity in the media and its close association with popular culture and celebrity. Therefore, by the time a SRC mitigation product or intervention is introduced to an athlete, they may already have pre-formed opinions on its effectiveness, and SRCs in general. This may be shaped by a combination of previous experiences, social observations, marketing, and media exposure, or trust in those recommending/implementing the intervention (Fig. 3) [57].

The reasons athletes adopt SRC mitigation interventions have not been elucidated through research. Often, they are mandated to comply with certain interventions by an individual, sports organization, or governing body, or they could also be assigned as part of a research study. However, many choose to purchase these products because of their belief in them and then report feeling an actual benefit and recommend the product to others. Numerous factors ultimately shape these positive expectations.

5.1 Marketing and Media Exposure

Patient expectation can be a key mediator of placebo [58]. Numerous products have claimed to reduce SRC risk, including helmets, mouthguards, headbands, and artificial turf surfaces. These claims are often not supported by high-quality epidemiology research, as evidenced by several decisions from the US Federal Trade Commission in its crackdown on inappropriate marketing for SRC [59–62]. Regardless of the accuracy of these claims, consumers are still exposed to product advertising and marketing, which can influence expectations for effectiveness [63].

The Q-collar jugular compression device provides an excellent example of how media exposure might shape expectations, even at the research stage. The original claim was that the Q-collar reduces SRC risk by mimicking the physiological adaptations of woodpeckers. The company has since shifted the narrative away from SRC risk, in favor of more generic "brain protection", and the woodpecker justification has been debunked [64]. Nonetheless, a campaign announcing the Q-collar as a possible SRC preventive measure launched in 2014 [65], with numerous mainstream media stories making bold claims about how it could prevent concussions [66–72]. The research trials examining the Q-collar were profiled by local news media, with participants commenting on their excitement at being involved [73] and filmed during their laboratory visits [74, 75]. News stories had reported the device was previously shown to be effective in other sports [76, 77] and the

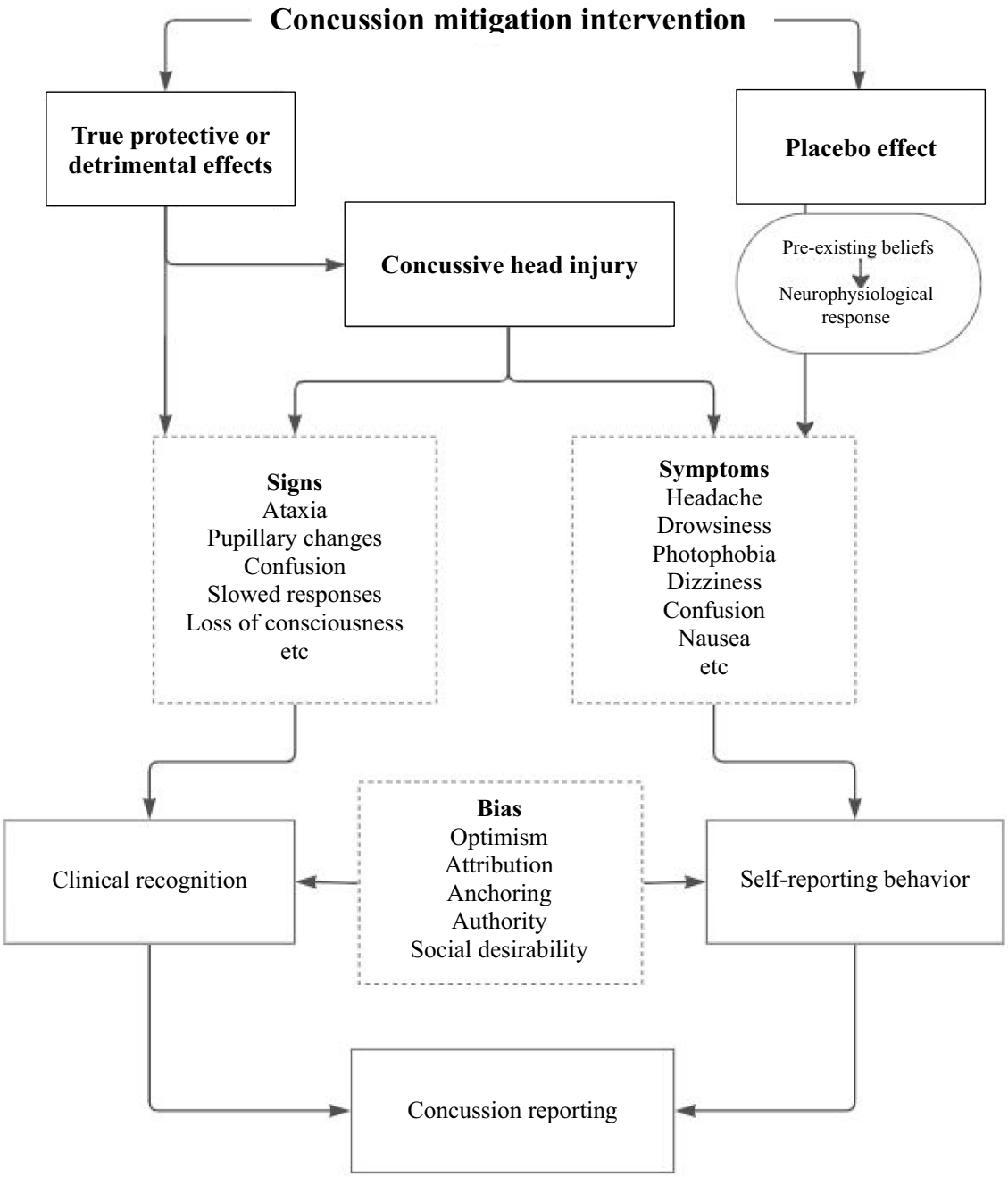

Fig. 2 Model of how concussion prevention interventions ultimately influence the number of reported concussions. True protective effects and placebo effects may each play a role in influencing an athlete's clinical signs and symptoms. The actual numbers of concussions reported ultimately reflect a balance between these factors. Symptoms themselves may be reduced, absent, or delayed because of a placebo effect, which could allow athletes to pass an initial sports-related concussion (SRC) screening test. Clinical recognition of SRC signs may be influenced by a healthcare worker's biases about the effectiveness of an SRC intervention. Likewise, self-reporting of acute or delayed SRC symptoms may be influenced by an athlete's bias. These are not limited to placebo effects (i.e., reduced symptoms), but also include misattribution (e.g., belief headache is due to dehydration, not SRC) and intentional non-disclosure (e.g., motivations to conceal symptoms). If an intervention has no protective effect, but a modest placebo effect, one would expect it to produce fewer SRCs in a randomized controlled trial, and thus produce a spuriously positive finding

study's Principal Investigator stated "*if this is right, this will be the single most [game] changer in sport*" [76, 77]. Local news stories about their participation stated "*there is plenty of reason for optimism*" [78] and the public relations director for one school stated the participants "*feel pretty honored to be part of this*." Thus, it is difficult

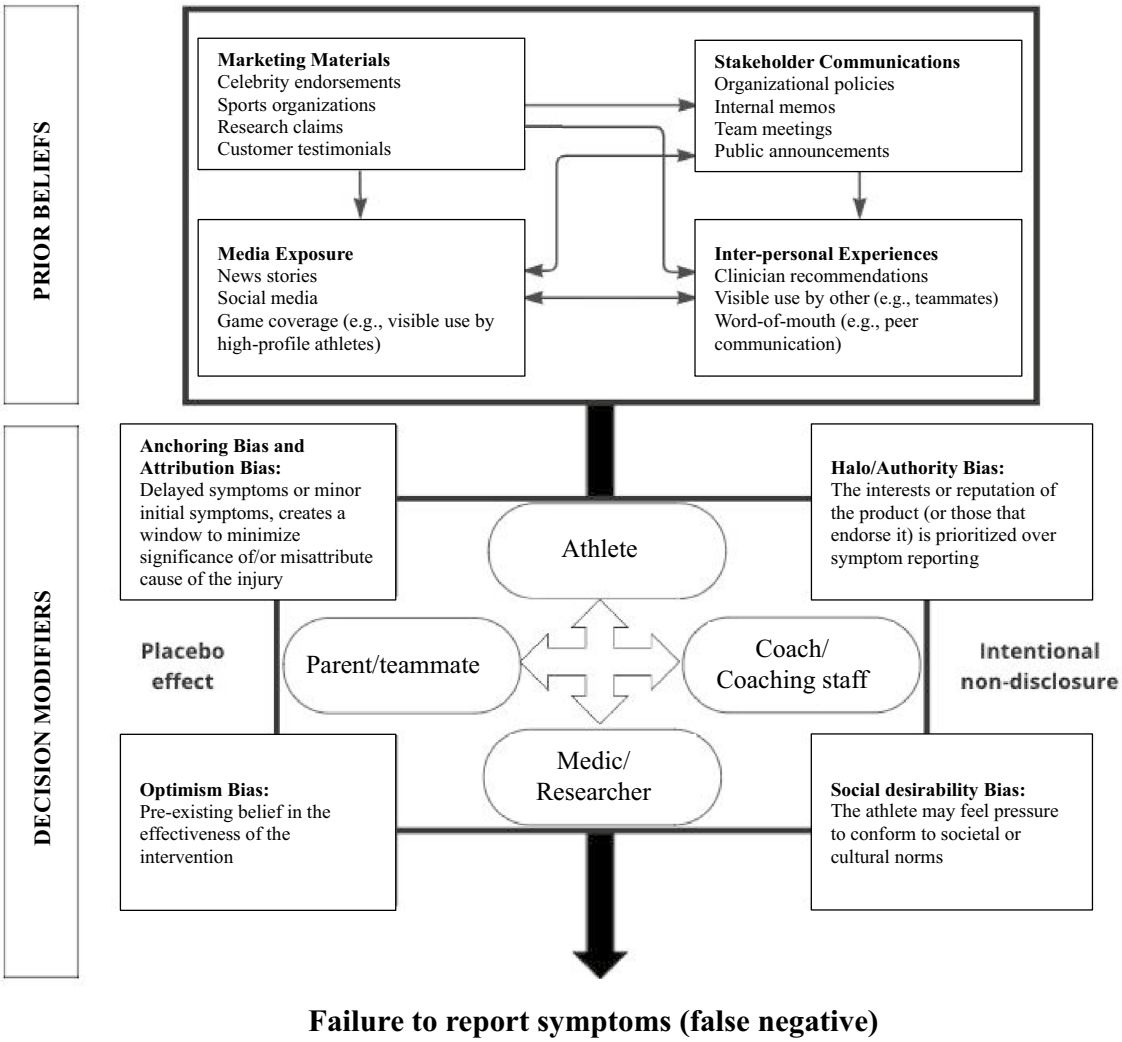

Fig. 3 A model of how prior beliefs and decision modifiers may interact to ultimately influence concussion reporting. Numerous factors can influence prior beliefs about a sports-related concussion intervention, which then influence behaviors. Prior beliefs about intervention effectiveness may arise from various forms of exposure. Each stakeholder is potentially subject to bias, and these stakeholders influence one another (e.g., a parent's optimism bias may influence the athlete's personal beliefs). We posit that the net result is that a prior belief in the effectiveness of sports-related concussion interventions ultimately leads to fewer sports-related concussion being reported, both through placebo effects and intentional non-disclosure

to imagine that athletes assigned to the collar intervention did not have some pre-conceived notion or desire for the device to be effective.

5.2 Organizational Trust

While there is no research specifically examining how sports organization support (or lack thereof) influences perceptions on the effectiveness of an SRC intervention, padded helmet shells provide some high-profile anecdotal examples. In 2022, numerous NFL players began wearing the Guardian Cap (a soft-shell cover that fits over the hard-shell American Football helmet) and the NFL provided some summary statistics suggesting the device reduced concussion risk (study methodology and raw data were not provided) [79]. According to NFL press releases, the device reduced concussion risk by 30% in a selected subset of positions who were mandated to use the device. The NFL later announced it would expand its mandate of the device [80]. Given the significant status and branding of the NFL, it is likely that the public (including players and coaches) will interpret their policies or actions in high regard.

Such a positive endorsement can influence behavior in other populations and key stakeholders. After the NFL press release, some high school coaches supported the mandatory use of these devices [80], with one quoted as

saying, "*if they didn't work and they didn't keep people safer, I really don't think the NFL would be mandating what they are*." This demonstrates a form of unconscious cognitive bias that aligns with a halo effect [81]. In this case, the magnitude and scale of the NFL's brand can influence the public's judgment, making them more likely to unthinkingly follow their policies or advice, rather than making an independent judgment based on (in this example) the study's methodological rigor and data transparency.

5.3 Celebrity Endorsement

New healthcare interventions are often marketed through celebrity athlete use and endorsement [82]. Persons in the public eye, and more recently the rise of influencers, can dramatically shape public discourse and consumer behavior [83]. For example, education entertainment is a contemporary genre of health promotion where content is strategically embedded into popular media streams (e.g., social media, music, television) to impart knowledge and influence behavior in young people. However, its relevance and success ultimately depend on the quality and fidelity of the educational content.

A number of SRC mitigation/recovery products have been endorsed by various sporting celebrities (e.g., Neurosafe, Tom Brady [84]; recovery water, Russell Wilson [85]). We acknowledge that celebrity endorsements do have general benefits including: building an emotional association with the product, boosting long-term awareness of an issue, increasing trust for a product particularly if consumers have no prior knowledge of the brand or product itself, and opening up new markets (in this case, a wider variety of sports) [86]. However, a concern is that these products were not supported by robust empirical data [30], which gets lost in the wider conversation about these products.

Celebrity use also provides social proof of effectiveness. In the case of the Q-collar, Luke Kuechly and Vernon Davis were among the first to use the device in the NFL, and their adoption of it received significant media attention. Even though both were concussed shortly after they began wearing the Q-collar [87, 88], media stories discussed how the athletes believe they benefitted from it [32, 33]. Both now work for the company promoting the device. Other sports celebrities have spoken out about how they use it because they value their brain health. These messages are a "double-edged sword," in that they do create awareness about the importance of long-term brain health, yet they also persuade consumers to embrace products rooted in poor evidence. While these athletes seemingly are not scientists who understand the major problems with the device and the research behind it, including unrealistic physiological mechanisms [64, 89] and a biased interpretation of clinical trial data [90], they may serve as "proof" that it must be effective. Importantly, we are not aware of any instances of celebrity athletes stating that specific SRC prevention products are ineffective, nor are we aware of athletes encouraging consumers to be skeptical about the validity of such products.

6 Plausibility of Placebo Effect for SRC Mitigation

We are not aware of any research that has specifically examined placebo effects in SRC mitigation (or that prevented any other acute sports injuries). However, one can gain insight from prophylaxis studies for other conditions, many of which have symptoms that overlap with SRC.

6.1 Migraine Prophylaxis

Migraine diagnosis is based on a wide array of self-report symptoms (e.g. headache, aura, photophobia, nausea), many of which overlap with concussion. Treatment or prevention efficacy in migraine research is commonly evaluated by comparing the differential responses of treatment and placebo arms, using a parallel randomized design. A common assumption is that any positive clinical effects are due to the additive effect of active/specific effects of the treatment and contextual effects (e.g., natural history, regression to mean, placebo). However, recent research suggests that the relationship between specific and contextual effects of migraine prophylaxis is more complex.

In an analysis of 71 experimental studies, Tepper et al. [91] found that the magnitude of the placebo response in preventative migraine interventions is not stable, having increased significantly in the past 20 years. This pattern was largely explained by the increased use of injectable or intravenous placebos in research, rather than its oral administration. The complexity of placebo and its influence on self-reported migraine symptoms is also evident in a recent meta-analysis, whereby sham acupuncture and sham surgery produced comparable effects to active oral pharmacological therapy, and oral "pharmacological" placebo was no better than no treatment [92]. Similar placebo effects are seen in cluster headache prophylaxis [93]. Regardless, the ability of certain placebos to prevent the onset of migraines lends support to the notion that a placebo could also stop or delay SRC symptoms.

6.2 Audiovestibular Conditions

Sports-related concussion often involves audiovestibular symptoms, and these too are sensitive to placebo effects [94].

6.2.1 Motion Sickness

Motion sickness prophylaxis provides an excellent example of sensitivity to placebo, as it can be experimentally induced in a laboratory setting through nauseogenic body rotation. This readily allows for the effectiveness of attenuation measures (and placebo effects) to be examined. Horing et al. [95] used a balanced placebo design to determine the magnitude to which a placebo influenced symptoms and objective physiological measurements of motion sickness. They found that symptoms of motion sickness improved when participants were deceived into thinking a placebo was an anti-emetic agent (i.e., a drug known to reduce motion sickness). Importantly, physiological responses to the nauseogenic body rotation (validated gastric electrical measurements) were not responsive to this placebo effect. In a similar study, Jacob et al. [96] found that placebo "treatment" was more effective than no treatment in reducing nausea and symptoms of motion sickness following a different nauseogenic protocol. They also experimentally manipulated stress levels and found that physiologic sequelae (gastric activity) were actually improved with placebo under the non-stressed condition, but not the stressed condition. These studies demonstrate how a placebo can influence a patient's symptoms without altering the underlying physiological response.

6.2.2 Tinnitus

Research examining tinnitus (ear ringing) also seems sensitive to placebo effects. For instance, a 1984 trial [97] recruited 20 patients with tinnitus who had received placebo treatment in a previous study, who were then told they would receive a potentially effective treatment in this new study but that their individual response could vary. However, this was a deception study, and participants were injected with saline. Six participants (30%) reported significant improvements in tinnitus (placebo effect), two (10%) reported significant worsening (nocebo effect), and the remainder reported no change following the sham injection [97].

6.2.3 Meniere's Disease

Placebos seem to be effective in managing symptoms of Meniere's disease, a condition that includes tinnitus, hearing changes, and vertigo. A 2024 meta-analysis reported improvement in short- and long-term subjective patient symptoms, including a 52.5% reduction in vertigo attacks [98]. It must be noted that the experimental group did achieve superior results (90.1% improvement), but the authors noted that the placebo effect was still "impressive." One of the trials included in the study even reported a sham procedure (mastoidectomy) was equally effective as the treatment procedure (endolymphatic shunt) at the 9-year follow-up [99].

7 Implications of a Placebo Effect for SRC Mitigation

If a placebo effect is a major mechanism in the apparent effectiveness of SRC mitigation interventions, this poses various ethical questions. As we have already mentioned, there is limited research examining placebo effects for preventing acute injuries or new-onset disease, thus we cannot readily extrapolate existing ethical debates about placebo effects to SRC.

It has been argued that placebos do have an important place in medicine, as they may allow patients to feel better and improve quality of life. [100] Some studies even demonstrate that participants who have greater adherence to their assigned placebo actually have better clinical outcomes (e.g., longer survival time in cardiac patients) [101]. Thus, it could be argued that if an SRC intervention produces a placebo effect, this may ultimately be beneficial to the athlete. However, we posit that placebo effects in SRC mitigation would ultimately cause more damage than benefit, for the reasons presented throughout this section.

7.1 Underlying Neuropathology Not Addressed

If a placebo effect is present in SRC mitigation, it likely represents altered symptoms and perceptions, but not necessarily reduced neuropathology. Examples of the disconnect between symptoms and actual pathology are described in the placebo literature, but perhaps best exemplified by asthma research. A double-blinded crossover study demonstrated that an albuterol inhaler (active drug), placebo inhaler, and sham acupuncture all reduced symptoms of asthma by approximately 45–50% compared with no treatment, though only albuterol administration actually improved the physiologic measurement of airway function (forced expiratory volume in 1 s: 20% improvement in the albuterol group, 7% improvement in all other groups) [102]. Similar examples of changes in symptoms without changes in underlying pathophysiology can be found elsewhere in the literature. Herein lies the danger of the placebo effect in prophylaxis—participants may experience fewer SRC symptoms, yet continue to expose themselves to further injurious stimuli while underlying neuropathology remains unresolved.

Thus, placebo may play a role in athlete concealment of concussion and could increase the likelihood of continuing to play (or prematurely returning to play) while having acute neurologic damage [103, 104]. Concussion causes a neurometabolic cascade that leaves

the brain in a vulnerable state for days to weeks. This period is characterized by ionic shifts, excitotoxicity, and energy crisis [103]. Further injuries during this time can exacerbate physiological disruption and delay healing. If athletes, who are already hesitant to self-report SRC, are further influenced to continue playing by equipment and devices that are no better than placebo, our concern is that greater neurological damage may occur, which may only become evident over time. Athletes receiving multiple SRCs have raised concerns given the increased risks of musculoskeletal injuries [105, 106] but also the longer term consequences on cognitive health following multiple SRCs. [107] Emerging evidence suggests that individuals who experience multiple SRCs are at risk of later-life cognitive impairments [107], and also face an increased probability of developing neurodegenerative diseases due to exposure to neurotrauma [108].

7.2 Erroneous Conflation of SRC Protection with a Long-Term Risk

Placebo effects may also provide a false sense of safety and influence decisions about collision sport participation. "Advances in equipment safety and helmet technology" has been identified as a factor that reduces parental concern in youth American-style football [109] and influences participation decisions [110]. More research is needed in this realm, but customer reviews of SRC products do indicate a sense of reduced risk influencing participation, including the mitigation of long-term risks. However, this is not rooted in any evidence.

While the debate centers around SRC prevention, recent evidence demonstrates that the risk of CTE is not due to concussions per se, but rather from the cumulative exposure of repetitive neurotrauma [111]. Thus, it is important to recognize that interventions that mitigate short-term signs and symptoms of SRC do not necessarily translate to long-term CTE risk reduction. Those who conflate improvement in short-term symptoms with true long-term neurologic protection may have unrealistic expectations regarding their CTE or traumatic encephalopathy syndrome (TES) risk.

Attempts to reduce the CTE risk through equipment or devices have been ineffective; for example, many cases of CTE have been in athletes who have worn helmets (both hard-shell and soft-shell). To date, the only known risk mitigation from CTE is reduced exposure [108]. The CTE mitigation literature is limited because of the long-term pathogenesis of disease and the inability to diagnose it in living patients. Thus, TES is a relevant clinical endpoint to consider, as it is a clinical diagnosis [112, 113]. We are not aware of any literature examining interventions to mitigate TES.

7.2.1 Long-Term Placebo Response

We can only theorize on whether the placebo effect could influence TES. We would expect that the placebo effect would be greatest while an athlete is actively using an SRC intervention. In the case of protective equipment and rule changes, we would expect the athlete to experience a neurobiologic response (i.e., dopamine release, activation of neural reward circuity) each time the intervention is employed. This event-based physiological response would cease to be activated when the intervention is terminated (e.g., voluntary discontinuation, termination of research study, retirement from contact sport). We would expect that any residual neurobiologic placebo response that could attenuate TES symptoms would also dissipate with time.

We may gain some insight into lingering long-term placebo effects by evaluating studies involving a single placebo treatment, such as sham surgeries for orthopedic conditions. These studies do often report placebo effects that last for 12–24 months, and one study found similar clinical outcomes 5 years after sham surgery. [114] However, these studies also often have several limitations that undermine their internal validity [115], and the "real" treatments may actually cause harm [114]. Likewise, a 3-week intervention for back pain found the placebo group still had benefits 5 years after study termination [116], though it is difficult to differentiate between placebo effect and natural history. These studies demonstrate the challenges in determining how long a placebo effect can last following termination of an intervention.

7.2.2 Challenges of Examining Placebo Effects in CTE and TES

It is our opinion that athletes who sustain neurological trauma from repeated head impacts will not experience long-term symptom relief from SRC interventions used during their playing career. Any perceived benefits from these interventions, likely attributable to placebo effects, are unlikely to persist years or decades after they have stopped playing. However, this would be challenging to convincingly demonstrate or refute.

Chronic traumatic encephalopathy and TES are generally reported in older athletes after retirement from sport [113], but there are examples of athletes in their teens and twenties diagnosed with these conditions [117]. Regardless, it would require a multi-year study period to allow sufficient time for disease/symptom progression. Studies examining a SRC risk reduction are already challenging to carry out, and those examining interventions to reduce CTE and TES risks would be even more so for this reason. As cumulative impact exposure is a strong predictor of CTE risk (and likely TES risk), an ideal intervention would begin at the onset

of contact sports participation and follow these athletes throughout their career. Likewise, retrospective cohort studies would need to account for numerous confounders, some of which would be nearly impossible to quantify (e.g., helmet type(s) and frequency of replacement, use of other protective interventions, quality of team medical supervision/care, field surfaces) and also subject to recall bias.

7.3 Risk Compensation

A common argument raised in SRC mitigation is risk compensation. This occurs when a proposed safety improvement (e.g., adoption of head gear) is offset by a change in behavior (e.g., more aggressive on field behavior) [118]. Currently the research regarding risk compensation is mixed and also sport/physical activity specific. For example, a 2019 systematic review investigating risk compensation in cycling reported 18 of 23 studies did not support the risk compensation hypothesis with regard to wearing of helmets [119]. However, more recent studies published after this systematic review have reported riskier behavior in bike riding when wearing a helmet [120, 121]. Conversely, in sports where concussion mitigation has become a major issue, such as Australian Rules football, the wearing of helmets has not appeared to have changed players' behaviors [122].

8 Limitations demonstrate Current SRC Mitigation Research

A 2023 systematic review and meta-analysis shows that SRC mitigation research typically involves a comparison between an intervention group and a control group, rather than with a placebo or sham group [1]. Only 6.3% (12/192) of comparative studies in this field used a randomized controlled design and only seven of those studies recorded concussion diagnoses (as opposed to surrogate measures) [1]. Only two [123, 124] of the 12 RCTs used the term "single blinded"—the remainder did not provide any information regarding participant blinding, placebo, or how the interventions (e.g., helmets, mouthguards) were framed to the participants. Only one RCT reported assessor blinding, but these were for a surrogate outcome (magnetic resonance imaging parameters), not SRC [125]. Trials also did not mention blinded statistical analyses, and none mentioned or suggested the possibility of a placebo effect. Despite these shortcomings, 96% of studies were classified with either "acceptable" or "excellent" methodological quality and there were no cases of serious risk of bias highlighted as part of their GRADE recommendations. Thus, the potential for systematic bias because of a lack of blinding/placebos does not seem to be perceived as a major concern for the integrity of SRC research, but this nonchalance is not justified.

We acknowledge that it is challenging to blind all individuals in SRC mitigation research, but we cannot ignore the biases that arise when these elements are omitted from the RCT design. Athletes and medical staff can see SRC mitigation devices (e.g., soft-shell helmet add-ons, neck collars), and are informed of rule changes designed to reduce risk. Awareness of interventions, combined with preconceived expectations, may subconsciously affect behaviors (observer bias), perceptions of SRC risk, symptoms, and subsequent medical evaluations. It is reasonable to suggest that some of the non-specific effects alleviating migraine symptoms and other neurological conditions (described in Sect. 6) could also hold true for SRC mitigation.

Currently, the proportion of observed effects attributable to "specific" versus "non-specific" (placebo and other factors) effects in SRC research is unclear, and we know little about their potential interaction. It is possible that the contribution of placebo varies based on contextual factors such as athlete expectation, or the sporting/clinical medium through which the intervention is delivered. Examining commonly used mitigation strategies based on a standard two-group parallel design (e.g., treatment vs placebo) could help to disentangle some of the specific versus non-specific effects. To our knowledge, there are currently no such studies in this field. However, to fully understand the placebo response, we must also consider key contextual effects moderating the specific treatment effect; this would require further examination of treatment efficacy again using explanatory, placebo-controlled, or sham-controlled designs, but where there are at least two "intensities" of placebo effects.

9 Recommendations for SRC Mitigation Research

9.1 Study Design

Identifying specific and non-specific effects in SRC interventions and their interactions may be quite complex. An example of a balanced placebo design in this field could involve four distinct conditions:

- intervention group (participants are given information on expected benefits);
- intervention group (no information provided on expected benefits);
- placebo/sham group (participants are given information on "expected benefits"; requires a deception component);
- placebo/sham group (no information on "expected benefits").

Central to this design is that participants in the intervention groups are provided with identical equipment, but its description is intentionally altered to influence patient expectations [57]. For example, if the intervention of interest is a novel helmet design (e.g., one with a new padding material designed to reduce shock transmission), half of those allocated to receive this would be informed it is highly effective in reducing SRC risk, while the remainder would be told it has a different unrelated purpose (e.g., reducing methicillin-resistant *Staphylococcus aureus* skin infection risk). Members of the sham group would all be provided with identical looking helmets to the intervention group, but these would lack the novel protective mechanism; the deception component involves half of the sham group being informed that their helmet has new SRC mitigation technology. This design helps to disentangle the physiological effect of the intervention from the psychological effects of expectations, beliefs, or biases.

Numerous ethical considerations must be addressed for this approach to SRC research. Deception is controversial for clinical research, but can be justified in certain situations and performed ethically [126]. Deception studies should include an authorized deception during informed consent (i.e., stating a deception will occur, but without specifics), and they must also debrief participants after the study's completion and measure the impact of deception on participants [126]. A further justification for ethical deception is that athletes and stakeholders may already be being deceived by unscrupulous marketing claims [82] and exposed to other experimental SRC mitigation interventions with unknown effects (e.g., rule changes). For instance, a policy change intended to reduce SRC risk in rugby actually increased risk [127, 128].

Athletes currently participating in high-risk sports are already consenting to be exposed to potential injurious stimuli (i.e., substantial efforts have been made to educate these populations about the risks of brain injuries) [42]. However, studies examining placebo effects in SRC mitigation interventions must not introduce new risks beyond existing sports or research participation. For instance, athletes in unblinded intervention studies may feel more protected and take extra risks, but sham studies seem unlikely to exacerbate existing risk compensation beyond that of traditional unblinded SRC studies. Likewise, the sham interventions must be fully compliant with existing safety standards (i.e., no existing level of protection should be taken away). This is a key difference from ethical debate over placebo-controlled drug trials in clinical populations—the proposed SRC study designs are not preventing research participants (i.e., athletes in high-risk sports) from receiving the current standard of care.

9.2 Challenges for SRC Studies

Identifying placebo effects in SRC interventions using the proposed methods will require more research participants. Increasing the number of groups, as described above, is necessary, but also the sample size in each group must be sufficiently high to achieve statistical power, especially if small-to-moderate effect sizes are anticipated. All of this increases financial costs for performing a study.

Design/production of appropriate sham devices will also likely increase costs beyond studies that provided only one intervention arm. The quality of the sham devices would also need to be independently assessed (i.e. investigators cannot have any current or even prior affiliations to these companies) to ensure they are not intentionally biased to perform worse than standard equipment (to help the experimental product being tested perform comparatively better). For example, unscrupulous helmet companies could potentially provide identical-looking sham helmets, but with intentionally compromised shell materials that reduce injury resistance. Though this may seem far-fetched, there are examples of pharmaceutical companies comparing their new product to insufficient dosages of existing therapies or having placebos that have substances that increase the risk of side effects [129].

Providing different expectations and the use of sham devices is challenging. Experimental rigor requires that athletes do not share what information they received about their respective intervention with teammates, coaches, or medical staff. One option for reducing the likelihood of contamination is a clustered design, where the team is the unit of analysis, instead of the individual. These designs typically require larger samples (to account for an intra-cluster correlation coefficient), appropriate statistical methods (e.g., hierarchical or mixed-effect models), and stratified randomization to account for team-level confounders (e.g., differences in impact exposure, sociocultural factors, and clinicians evaluating suspected SRCs).

We would also anticipate resistance from SRC-prevention companies for funding such studies. Not only are we suggesting that SRC-prevention interventions may not be as effective as originally thought because of the placebo effects, but that future research in this field requires more financial resources. Clearly, these are not favorable conditions for industry-sponsored studies. Likewise, the extra costs for proper study design can create a financial barrier for smaller start-up companies to demonstrate the effectiveness of novel equipment. Thus, larger organizational grants may be needed to support this updated approach to SRC research. However, we believe that the increased cost of SRC-prevention research will be outweighed by higher quality evidence regarding a SRC intervention's effectiveness.

It must be acknowledged that these recommendations are easier to implement for SRC mitigation products, compared with a policy-related (e.g., rule changes) or technique-based (e.g., lower tackle height) intervention. However, this does not mean such effects are impossible to include. For instance, if a sports organization has reason to believe a new policy or technique will specifically reduce SRC risk, they can implement this rule without specifically stating this rationale, as doing so might introduce biases (and influence reporting and recognition). Instead, an alternate rationale could be provided (e.g., stating the rule is intended to reduce anterior cruciate ligament injury risk, to facilitate officiating, or to enhance the entertainment value), and the SRC rate could then be monitored. If a significant reduction in SRC was observed, one could have more confidence this was due to the rule itself, rather than social desirability bias.

In accordance with all clinical trials, each of these research designs would benefit from pre-registration in a public database. While this creates a theoretical risk that participants become aware of the deceptive components of the research, this is unlikely. Furthermore, any such concerns are outweighed by the benefits of pre-registration; particularly, that it ensures published results can be compared to the pre-registration protocol, and reduces the risk of post-hoc changes to study protocols or analyses to favor desired outcomes.

10 Conclusions

Sports-related concussion mitigation interventions receive substantial attention from researchers, clinicians, mainstream media, and athletes/sports organizations, yet the roles of placebo and other non-specific effects remain largely unexplored. As the diagnosis of is currently dependent upon subjective clinical diagnoses based on signs/symptom presentation, in a population who may be ignorant to recognize, or willful to misattribute/hide signs/symptoms, it is important that clinical trials are designed to rigorously mitigate potential placebo effects.

Inflated perceptions of SRC mitigation effectiveness may facilitate athletes' continued participation while concussed, or influence decisions to participate in high-risk sports. Thus, there is a need for high-quality independent research to differentiate between specific versus placebo effects in SRC mitigation interventions. While numerous challenges exist to improve the quality of this field of research, we opine that it is not appropriate to dismiss them as too challenging or costly to implement. Either SRC-prevention studies must be appropriately designed to differentiate actual protective effects from placebo effects or they should explicitly acknowledge the possibility that any observed benefits are due to placebo effects (rather than actual protective effects). This level of rigor in the SRC mitigation literature is necessary to ensure that decisions and policies intended to improve player safety and brain health are rooted in strong evidence of actual protection, rather than placebo effects.

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